



Evolving components: Haswell and GPUs



In recent memory, the term “post PC era” has been thrown around a lot (even within the pages of this magazine) to signal the demise of personal computing as we grew up to know. Not for lack of thought or ability to weave words, mind you, but for the simple fact that the term's true. And no one's more worried with the impending doom of desktops, laptops, and ultra-mobile PCs more than the company that's built an empire manufacturing chips that power and run these devices. Intel. Intel deserves a lot of credit for accelerating the transition of desktop PCs into portable laptops, the microarchitecture they designed – increasing core performance while maintaining power efficiency – allowed laptops to shrink in size, be cool and become the primary computer for millions of users for close to a decade. But the world changed forever as Apple, Google and Samsung

embarked on the smartphone and tablet road. Intel was caught napping with netbooks, while little-known ARM (and its RISC-based architecture rivaling Intel) ushered in a revolution of epic proportions. And it's an unstoppable juggernaut. Experts believe that smartphone processors will be faster than current Ivy Bridge chips in less than a decade, and Intel has its work cut out to get a strong foothold in the mobile chipsets market, as ARM-based SoCs are pretty much the name of the game right now. AMD is no longer the only enemy, Atom isn't good enough and frankly has lost the plot, Intel needs to bring out the big guns if it has any shot at redeeming itself.

We may be in the post PC era, but in the tectonic shift between x86 and ARM, and the battle for hardware platform dominance, the legacy of Intel can't be written off that easily -- just because people are buying smartphones and tablets doesn't

mean they've stopped using their desktop or laptop PCs. As announced by Intel, 2013 is going to be the year of a whole new microarchitecture from the famous chip-maker, its next big tock (read more about Intel's Tick-Tock here: http://en.wikipedia.org/wiki/Intel_Tick-Tock). It's time for the 4th generation of Intel Core processors to rise and shine. Haswell, the successor to last year's Ivy Bridge and a much-improved hardware platform with a 22nm fabrication process which optimizes everything, is Intel's immediate answer to the assault from ARM-based hardware. Haswell claims to improve the performance and battery life of laptops, ultrabooks and hybrid devices to a whole new level, and truly utilize all the goodness of its 3D tri-gate transistors. The power saving on Haswell has improved by leaps and bounds compared to Sandy Bridge and Ivy Bridge chips, and so has the onboard GPU performance, with Intel claiming to offer up to 200% better graphics performance compared to Intel HD 4000 – now that's quite a feat. According to Intel, this year we will finally see an ultraportable laptop or hybrid device that can last for more than 10 hours on a single charge, thanks to Haswell chips, where Intel not only controls the power states of chips that it manufactures but also enlarges and optimizes the power saving envelope to third-party hardware (ICs, capacitors, etc) present on the rest of the board, stuff that otherwise isn't regulated and cracked down upon by Intel in the past. Intel's also ensuring that all Haswell-powered PCs to launch this year will have a minimum performance threshold that upholds its various claims, and to achieve this it's working closer than ever before with OEMs who ultimately mass-produce and distribute Intel-based products for the end user. While the jury's still out on

Haswell 2013 - Graphics naming (Levels and SKUs)

GT3e*	Intel Iris Pro graphics 5200	---
GT3* (28W)	Intel Iris graphics 5100	---
GT3* (15W)	Intel HD graphics 5000	---
GT2*	Intel HD graphics 4200 / 4400 / 4600	Intel HD graphics P4600 / P4700
GT1*	Intel HD graphics	---

Note: *GT is how Intel referred to graphics levels in NDA roadmaps so far

There are a bunch of changes Intel's ringing in with Haswell, apart from just a brand new socket for motherboards – LGA 1150 for desktop boards, others for mobile chipsets. With its additional on-die graphics memory (in gigabytes, yes!), Intel has found reason to classify its various graphics offerings (rolling out soon for the desktop and laptop PC market) be classified in five different levels. As is clear from the chart above, you can pretty much benefit from the graphics power depending on your requirements (and budget). The number of devices (ultrabooks / laptops / hybrids / all-in-one PCs / other form factors) sporting Haswell parts is going to increase exponentially, as now we do not just have the various CPU configurations to work with, but more varied GPU offerings as well. It will be really interesting to see how the Iris and Iris Pro GPUs perform in real life against the entry and mid-range discrete GPUs. Some of these graphics benefits will eventually percolate down to handheld devices, and that's an encouraging thought.